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REVIEW ARTICLE

Intensive Review of Drones Detection and Tracking: Linear Kalman Filter Versus Nonlinear Regression, an Analysis Case

Raed Abu Zitar[1] · Amani Mohsen[2] · Amal ElFallah Seghrouchni[3,4] · Frederic Barbaresco[5] · Nidal A. Al-Dmour[6]

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Abstract

In this paper, an extensive review for objects and drones (AUVs) detection and tracking is presented. The article presents state of the art methods used in detection and tracking of drones with adequate analysis and comparisons summarizing the findings of the most recent research material in that field. The most famous technique used in drones tracking is Kalman Filters (KFs) in its different forms. The paper presents analysis and comparisons for drones tracking based on Linear Kalman Filters (LKF) compared to tracking using Nonlinear Polynomial Regression (NPR) techniques. Interesting findings reflect the need for both methods at different circumstances depending on the noise conditions of the measurements. On the other hand, many new methods such as Artificial Intelligence (AI) based techniques are recently used in drones detection and recognition. Detection methods could come separate or combined with tracking techniques. The work presents broad and deep literature review with critical analysis of most famous methods used in drones detection and tracking.

1 Introduction

Objects and drones tracking in particular is drawing a lot of attention nowadays by wide sector of private and public entities. The drones are being used now in many aspects of life including, business, surveillance, research and military. The two issues of detection and tracking are inseparable for the drones regardless of their types or characteristics. The drones are usually small, fly with low altitudes, and can exhibit maneuvers in narrow vicinity of space. Moreover, it can be easily mixed with birds and other flying objects, especially in complex environments where clutters and noise are inevitable. This is the main reason that we need to detect and recognize before we track, or detection and tracking can come together in order to avoid delays and make use of the information that the tracks have in detection and recognition (for example to decide if the object is a drone or a bird). There is wide variety of detection and tracking techniques that vary in approach and technique. All relay on high quality sensors (radars or cameras) with appropriate motion and measurement models, followed by efficient estimation mechanism that handles uncertainty interactively.

The structure of the paper is divided as follows; the second section is a literature review for most relevant techniques of drones detection, the third section is a relevant review for drones tracking techniques, followed by state of the art that combines detection with tracking. Section five is covering the most important and vastly used techniques of Kalman Filter (KF), section six is introducing the Polynomial Nonlinear Regression (PNR) method in drones tracking opposed to (Linaer Kalman Filter) LKF tracking. Finally, a section for conclusions and discussions is presented.

✉ Raed Abu Zitar
raed.zitar@sorbonne.ae

Amani Mohsen
A00021501@sorbonne.ae

Amal ElFallah Seghrouchni
Amal.Elfallah@lip6.fr

Frederic Barbaresco
frederic.barbaresco@thalesgroup.com

Nidal A. Al-Dmour
nidal@mutah.edu.jo

1 Thales/SCAI, Sorbonne University, Abu Dhabi, UAE

2 Department of Physics and Sciences, Sorbonne University, Abu Dhabi, UAE

3 Mohammed VI Polytechnic University-Ai movement, Rabat, Morocco

4 Sorbonne University, Paris, France

5 Thales Land and Air Systems, Paris, France

6 Department of Computer Engineering, Mutah University, Mutah, Jordan

The methodology in this paper can be summarized by starting with state of the art literature review for both detection and tracking that relies mostly on artificial Intelligence (AI) techniques. In particular, Deep Learning Neural Networks (DNN) and Convectional Neural Networks (CNN) are presented. Probabilistic and statistical techniques for detection and tracking are also introduced. Kalman Filters that are Bayes Theorem based techniques are also presented, expressing the success of those techniques that have been used since long time ago. On the other hand, a deterministic statistical method (PNR) is compare with KFs to demonstrate how those two methods can be compared. We try to cover two major types of sensors; cameras and radars, and compare between AI-based and classical machine learning methods. Finally, show analysis for stochastic (KF) and deterministic (PNR) methods. A solid simulation tool (stone soup) is used to simulate measurements and tracks for the analysis part. Stone Soup is An open framework in PYTHON that has been develop by US/CAN/UK/AUS MoDs https://stonesoup.readthedocs.Io/en/v0.1b5/auto_tutorials/index.html

It includes also an UAV tracking Radar Demonstration: https://stonesoup.readthedocs.Io/en/v0.1b5/auto_demos

The goal is to provide an easily comprehensible reference for state of the art drones detection and tracking.

2 Literature Review for Drones/Objects Detection Methods

We start by reviewing the state of the art of drone/object detection which relies mostly now on deep learning techniques. Below are some resources that are selected based on relevance:

1. *You Only Look Once (YOLO): Unified, Real-Time Object Detection*[59]. This paper proposes a novel approach that considerably reduce the computation complexity of real-time object detection compared to existing methods. These methods usually use sliding windows of various scales over which they perform classification. The more recent R-CNN (Region-based Convolutional Neural Network) generates bounding boxes before classifying. Their technique reformulates the problem as a regression; the image is divide in equal cells and for each cell a given number of bounding boxes are predicted with corresponding confidence scores, along with a probability score for each class (only one per cell and not per box). A simple neural network is used; 7 convolutional layers to extract features, they are followed by two fully connected layers which predict the classes and bounding boxes. However, the fact that only one object can be predicted for each cell limits the ability of YOLO to detect small objects and flocks.
2. *YOLO9000: Better, Faster, Stronger* [63]. In this second version of YOLO the authors propose a joint training algorithm which allows to extend the limited object detection data with image classification data. As a result the number of classes detected goes from 20 to 9000.
3. *YOLOv3: An Incremental Improvement*[64]. In this work, bounding boxes are predicted at 3 different scales. the bounding box priors are found using k-means as in the work preceding version. The feature extraction at different scales is inspired by [44]. It compares well with other architectures being only beaten by RetinaNet [45] for detection performance metric on COCO data set (although it is faster). This method still performs better when the metric is not so demanding in terms of bounding box accuracy (an IoU (Intersection over Union) threshold of up to 0.5). Code and demo are available at https://pjreddie.com/darknet/yolo/.
4. *YOLO-LITE: A Real-Time Object Detection Algorithm Optimized for Non-GPU Computers* [56]. In this paper, the authors aim at modifying YOLO (algorithm V2) [63] in order to make it usable in real time on non-GPU computers and hence practical for many applications. They divide the input image size by 2, remove some layers and remove batch normalization(the covariant shift and vanishing gradient problems addressed by batch normalization no longer arise here due to the smaller size of the network (as argued by the author) so removing it doesn't reduce performance). They manage to obtain a processing speed with a CPU above 10 FPS (Frames Per Second) while keeping the (mean Average Precision) mAP above 30% on PascalVoc [88] (30.24% VS. 40.48% for tiny YOLOV2). Aside from FPS processed by the model, metrics such as number of layers, number of filters and FLOPs are compared. Code and demo available at https://reu2018dl.github.io/.
5. *Rich feature hierarchies for accurate object detection and semantic segmentation* [18]. In this paper the well known object detection model R-CNN is introduced. It is structured as follows: regions proposal module, CNN module that extract features from each region, and SVM's for each class. For region proposal, selective search is used [86] (software is available at http://disi.unitn.it/~uijlings/MyHomepage/index.php#page=projects1). Before being fed to a CNN the regions are wrapped to the correct input size (227x227) and mean-subtracted. The CNN is pre-trained using the classification data set ILSVRC2012 [69], then fine tuned for the detection task. The features from each regions (4096 features for each of the 2000 images) are then multiplied with a Support Vector Machine(SVM) weights for each classes in order to perform classification (which

amounts to a matrix product for each image). Bounding boxes of detected objects are corrected using ridge regression on the features. This method achieved significantly better score than its competitors (mAP of 34.1% on ILSVRC2013 VS. 24.3% for Overfeat model [74], the second best result).Source code is available at http://www.rossgirshick. Info/.

6. *Pedestrian Detection with Unsupervised Multi-Stage Feature Learning*[75] In this paper, the authors propose to pre-train a convolutional network in an unsupervised fashioned to extract features before fine training it along with a classifier (SVM) for the task of detecting pedestrians (similar to drone detection). The presence of "layer skipping connections" allows to enforce multi-scale features. The unsupervised training is done hierarchically; each layer is trained using the output from the previous layer following the approach presented in [25]. Sparse coding (a linear reconstruction model) is used. Please, see [40] where, at each training step, a sparse representation of the input is found using an optimization algorithm, then an stochastic gradient update is performed on the parameters (coefficients, filters and bias of the convolutional layer, plus filters used to match the sparse representation with the input) in order to get closer to the sparse representation.
7. *Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks*[65]. Here, Region Proposal Networks (RPN's) are introduced allowing to gain a significant amount of time compared to the selective search method used in [18] (10ms VS. 2 s per image on a GPU). code is available at https://github.com/shaoqingren/faster_rcnn (in MATLAB) and https://github.com/rbgirshick/py-faster-rcnn. The architecture is as follows: a shared feature map is extracted using convolutional layers. for the bounding, a sliding window passes over the feature map and predicts *k* (pre-set scale and aspect ratio combinations) bounding boxes along with the correspoding 'objectness' scores through two separate fully connected layers. Then the classes are predicted using Fast R-CNN [17]. For the shared convolutional layers, two existing architectures are; the ZF [92] and the(much heavier) VGG-16 [78] are used. When trained on PASCAL VOC 2007 [28], the version with ZF achieves a mAP of 59.9% and 17fps, while the one with VGG-16 reaches a mAP of 69.9% at 5fps, in comparison the the previous version of Fast R-CNN, there is a gain in accuracy in addition to the substantial gain in time.
8. *SSD: Single Shot MultiBox Detector* [47] This paper presents a new neural network approach to object detection that significantly increases detection speed while maintaining accuracy compared to existing state-of-the-art approaches (Faster R-CNN [65] and YOLO [59]). As in YOLO, the time efficiency of this methods comes from avoiding to resample pixels or features once the bounding box has been estimated, but unlike YOLO it does not lose accuracy compared to state-of-the-art methods along the way, as summarized below[47]: They Use an approach similar to YOLO, except they use small convolutional filters applied to features map (at different stages to predict different scales) to predict box offsets and category scores. As in Faster R-CNN, VGG-16 network is used for the early convolutional layers.

3 Literature Review for Objects/Drones Tracking Methods

Object/drone tracking is the process of estimating the state space values of an object that is moving in an undetermined manner with or without the help of the sensors. It can be implemented in several methods, however, in this section, we present two state of the art *probabilistic* methods:

1. *Online Tracking and Reacquisition Using Co-trained Generative and Discriminative Trackers*[91] When tracking an object, the said object can change appearance due to variation of lighting and angle, be lost in a clutter or even become occluded. The goal of the paper is to define a model capable of learning the appearance of an object on the fly while tracking it, starting from limited labelled data, capable of reacquiring the object if it disappears. In this paper, the authors use co-training (*The basic idea is to train two classifiers on two conditionally independent views of the same data (with a small number of exemplars) and then use the prediction from each classifier to enlarge the training set of the other*) in order to combine the generative (joint distribution) and discriminating (conditional distribution) learning approaches. They use a Multiple Linear Subspace Generative Model, and an SVM (Support Vector Machine) classifier for the discriminative model, trained online with histogram of gradient. As new samples arrive, the subspaces are sequentially updated. When the number of subspaces have reached a maximum, the two most similar subspaces are merged. For the discriminative model, an incremental SVM algorithm LASVM [6] is used. In spite of very good results compared to existing literature, the authors point out that their model does not handle partial occlusion. They suggest that for tracking a particular type of object, combining offline training with online training would be beneficial as in [41, 43].
2. *Lévy State-Space Models for Tracking and Intent Prediction of Highly Maneuverable Objects*[16] Use a heavy

tailed α-stable Lévy process in a sate-space model for drone kinematic state and intent, which better captures sudden changes of direction typical with drones (as opposed to Gaussian process). The kinematic state and intent are enclosed in a vector that is modeled as the solution of an SDE (Stochastic Differential Equation) whose stochastic term (involving the Lévy process(es)) is calculated using a truncated Poisson series, while the other terms having explicit formulation. In the single α -stable noise source, the transition density is nicely formulated as a normal distribution. Similarly, Gaussian noise is added, except for the part of the stochastic integral related to the Gaussian noise that can be computed explicitly. Quick definitions of the Lévy models are as follows: Stable Lévy Langevin Model: used to model position and speed of an object with sensors. Stable Lévy Singer Model: models acceleration in addition. Stable Lévy Equilibrium Reverting Velocity Model: provides meta-level tracking (estimate intent position by modeling position and speed) Latent Destination Model: where the destination is modeled as a latent variable estimated by the model (as opposed to fixed vector in previous example). Algorithms for sampling these variables using observed data (in a Bayesian framework) are presented and tested.

4 Literature Review for Combined Drones Detection and Tracking Methods

This section focuses more on drone detection and tracking, either done separately or merged. There are many different techniques that vary also in precision, accuracy, and speed. The algorithms that are used in simultaneous detection and tracking usually have the following characteristics:

- Shape representation: point, primitive geometric shape, skeletal model, and silhouette
- Appearance representation: probability density, templates, multi-view appearance, active appearance model
- object detection methods are listed in two categories: single frame object methods, and methods using temporal information.
- For the tracking task the object detection and correspondence within the sequence of frames can either be done jointly or separately. Three family of tracking methods are described: point, kernel and silhouette tracking.
- Various tracking challenges such as cross camera tracking, treatment of occlusion, of non targets are then addressed through the presentation of a list of papers.

Several papers that tackle this subject are presented in the following subsections:

4.1 Robust Drone Classification Using Two-Stage Decision Trees and Results from SESAR SAFIR Trials [33]

Attributes and definitions presented in the paper:

1. *Context:* In the context of an increasing use of UAV's, non-cooperative drone surveillance is becoming more important. They can be small slow moving targets (low RCS), hence, with the use of staring radar, they can easily be confused with birds etc. Based on that, a classification model (a tree is proposed) to solve this detection/classification problem along side with the tracking process.
2. *Contribution:* uses both Kinematics and Micro-Doppler components for the classification. High -performance, real-time and INTEROPERABILITY.
3. *GameKeeper 16U:* The radar used in experimentation is built by Aveillant/ Thales.
4. *Target Features and Discrimination:* Kinematics and the basic features can be used to easily differentiate ground vehicles and high flying airborne from drones but birds have similar way to occupy air space hence the utility of Micro-Doppler signature (rotaries) is utilized.
5. *Classification a two stage approach:* Tree1: A priori knowledge, uses height to eliminate ground target and RCS (Radar Cross Section)and speed to eliminate Large and fast (non drone) ND targets. Tree2: trained with kinematic features and micro-Doppler information.
6. *Field Trials Results:* on the test data (false positive) FP ratio is more than halved for 2 stage (Drone Target) DT compared to 1 stage DT, which is important since typically and relatively large number of non-drone data points are processed.

4.2 The Application of Performance Metrics to Staring radar for Drone Surveillance [35]:

Attributes and definitions presented in the paper:

1. *Motivation:* overall system performance (tracking and detecting) not only classification (e.g. confusion matrix) are considered. The measures (metrics) presented could be extended to multi-sensor drone surveillance systems, they are based on the proposed metrics which encapsulates completeness, clarity, continuity, and kinematic accuracy.
2. *Proposing* the Single Integrated Air Picture (SIAP) Metrics: Rational and Definitions.
3. *Completeness*: average completeness of a target is the number of frames where at least one track is assigned to the target over the number of frames where the target

is detectable (active and inside the volume observed by the radar).

4. *Clarity in terms of ambiguity*: (of a target) instantaneous number of tracks assigned to the target (not defined if zero). The average is the average number of tracks assigned to the target over number of frames where at least one track is assigned to it.
5. *Clarity in terms of spuriousness* fraction (or number) of tracks that are not assigned to any target (regarded as false tracks).
6. *Continuity (rate of track ID changes)* measured in Hertz: difference between smallest number of tracks such that at least one track remains in frames where the target is detected and number of distinct periods (CPI) of tracking divided by the total duration of the tracking. It is averaged over targets-frame combinations.
7. *Kinematic accuracy: (of target j at frame k)* root of mean of square of differences between positions of tracks assigned to target j and true position target j. The average is calculated by giving equal weight to each target-frame-track combination.
8. Performance Results: Better results obtained when classifying the tracks using the two stage tree presented in previous article.

Precise formulas are given to compute these metrics. Data from the SESAR SAFIR programme are used to compute an example of performance results, before and after a classification step, showing a clear impact on the clarity metric. This data comes from live demonstrations performed in a combination of an urban and industrial environment close to the Port of Antwerp, Belgium. The radar system used is the L-band, 3-D, Aveillant Gamekeeper system [36] that has a 90° azimuth coverage and a detection range of 5 km for small drones.

4.3 A Comparison of Convolutional Object Detectors for Real-time Drone Tracking Using a PTZ Camera [54]

Attributes and definitions presented in the paper:

1. *Motivation:* Drones create a low cost overwhelming threat, hence there is need of low costs surveillance systems (cameras are much cheaper than radar surveillance systems). In this context, (Convolutional Neural Networks) CNN's can be utilized for drone detection. In this paper, a drone detection and tracking system using a Pan-Tilt-Zoom (PTZ) camera and an object detection algorithm is introduced. Six state-of-the-art CNN object detectors are compared.
2. *Study Approach:* Various existing object detection models are tested, adapted to only detect a drone class. The data set is made of 9525 labelled multi-rotor drones images taken from 1080p videos from different distances and angles, with different background, and 11 drone models of different size. Training set and testing set are separated in time in order to ensure different background and angles. 75% of drones have width smaller than 100 pixels (in images of width 1920).
3. *Drone Tracking System:* A Pan-Tilt-Zoom (PTZ) camera feeds images to the drone detection model, if a drone is identified a "pan-tilt-zoom" action is generated to track the drone.
4. *Study Results*

   - Accuracy: Considering both Precision and recall (F-measure) the best results are obtained with these architectures (starting from best) faster R-CNN with Inception Resnet, R-FCN with Resnet 101, and YOLOv2. In line with previous literature on small objects detection.
   - Speed (of trained detector): 3 best (in that order) SSD MobileNet, YOLOv2 and SSD Inception V2. Considering speed-accuracy trade-offs, YOLOv2 might be the most appropriate model in our usage.

4.4 Effective Ground-Truthing of Supervised Machine Learning for Drone Classification [76]

Attributes and definitions presented in the paper:

1. *Context:* With the increasing threat represented by cheap drones, there is an increasing demand for counter-UAS applications. Multi-beam staring radars are particularly effective (and "specifically designed") for detection of small targets as drones.
2. *Motivation:* Due to their high sensitivity, radars used for drone detection detect a high number of "confuser targets". Hence their is a need to correctly classify detected targets as drone or non-drone. As seen in [33] ground targets and air crafts can easily be differentiated from drones using for instance height and Radial velocity but the challenge comes from differentiating drones and birds. An effective combination of features allowing such classification, one feature being insufficient. Then a Machine Learning classifier should be trained, hence the need of labelled data coming from all the possible kinds of birds and possible "confuser targets". The goal of this paper is to introduce methods for collecting such ground truth data, and then train a model with these data.
3. *Methodology for ground-truths:* Using an application called "Theodolite" on an iPad to take picture of opportune targets along with meta-data registered by the app (the time, GPS position, altitude, bearing angle, eleva-

tion angle) that can be used to associate the target to the track recorded by the radar along with the type of bird that is manually entered by the person handling the tablet.

4. *Machine Learning Classification:* A decision tree is trained on the labelled data. It returns a binary output: drone or non-drone. the features used include height (and not other positional coordinates) derivatives of position, number of Micro-Doppler components, some statistical variable derived from features (mean, max, min), RCS and age of the track. The classifier results are good with around 95% of correctly classified data-points for both classes.

4.5 Multi-agent System to Design Next Generation of Airborne Platform[20]

Attributes and definitions presented in the paper:

1. *Motivation:* Utilize the increasing number and complexity of sensors onboard (i.e. the Remotely Piloted Aircraft System) by dynamically managing the resources allocated to them.
2. *Aim:* Design a Multi-Agent architecture for the real-time coordination of onboard
3. *The Mission Manager system (MSS) framework:* The MSS is the layer connecting the Mission Manager to all the sensors available. Its functions requires a high level of flexibility hence the suitability of Multi Agent System (MAS) architecture. In the proposed architecture sensors are permanent agents whereas transitory *tactical agents* are created when targets are detected (agentification). Each sensor agent follows a single *tactical agent* at a time.
4. *Scheduling:* The main contribution of this paper is to provide an efficient real-time scheduling approach for allowing the Mission Manager to collect as much information as possible about the ground and the state of the RPAS. This is done by assigning each *tactical agent* a priority level and asking them to require a task to the sensors then the scheduler allocates resources accordingly within a time frame and ask agents whose task has not been scheduled to resend a new task requiring less busy sensor.

5 Kalman Filter Integrated with Other Approaches for Target Detectiong and Tracking

One of the most important techniques used for moving objects tracking is Kalman Filter (KF) in its different types. The following subsections present a literature review for this approach incorporated with other methods that help in achieving better performance. Most of those methods are machine learning based methods:

5.1 A New Approach to Linear Filtering and Prediction Problems [39]

In this legacy paper, the Kalman filter is introduced as a solution to the so-called Wiener problem, it is interesting to follow the derivation of estimates of the transition matrix, filter applied to observed measurements, and covariance matrix of the estimation error for the optimal filtering presented.

5.2 Invariant Extended Kalman Filter (IEKF) Applied to Tracking for Air Traffic Control (ATC) [51]

This paper presents a new model for ATC with IEKF that aims at reducing the error in the estimated velocity vector during turns for small air crafts approaching airports. The existing algorithms are not well suited for this problem.

5.3 Interaction-Aware Kalman Neural Networks for Trajectory Prediction [38]

Interaction-aware Kalman Neural Networks (IaKNN) is a three layer architecture for autonomous vehicles trajectory planning. It is also a nine multi-agent framework prediction trajectory of multiple vehicles including the autonomous one. It has an Interaction aware acceleration which estimates a particular acceleration of surrounding objects that incorporate social interactions between them. These acceleration are fed to a classic kinematics model to produce interaction aware trajectories.

A Kalman filter layer is used to estimate trajectories from previous layer's output. This layer includes LSTM (Long Short Term Memory) to learn time varying measurement noises.

5.4 Cramér-Rao Bounds for Estimating Range, Velocity, and Direction with an Active Array [13]

It is Kalman Filter based model for estimating range and velocity of a target with the echo perceived by an array antenna is presented. The model aims at estimating range and velocity over using N samples taken over one Coherent Processing Interval (CPI). For this general model, Cramér-Rao Bounds are derived for the parameters, they are lower bounds on the variance of an estimator estimating a quantity from a samples, and allow to assess the quality of the estimator, the closer to the bound the better the estimator.

5.5 Angle-Only Tracking Filter in Modified Spherical Coordinates[79]

In this paper, authors extend the method of Kalman Filter tracking with modified polar coordinates (MPC) of [27] to 3D with modified spherical coordinates (MSC). As in [27], they argue that the inverse of the range is easier to observe due to its relation to with derivatives of angle measures which are directly observable.

5.6 Effective Ground-Truthing of Supervised Machine Learning for Drone Classification [77]

The article introduces of a new method to obtain ground truth labels for gamekeeper data. It uses a human observer on the ground to capture images of flying objects with a tablet and to provide a label for it (drone or bird). The GPS and the spatial measurement application "Theodolite" are then used to obtain a precise location for the objects. These ground-truth data can then be crossed with the radar data to obtain a data set that can be used to train a classifier. An example of decision tree classifier is trained with the "fictree" MATLAB function on this data set. It uses several extracted features such as height, velocity, acceleration, jerk, micro-Doppler components and target RCS (Radar Cross Section). It shows promising results with 94.3 % of correct classification. Kalman Filters are used in tracking.

This section covers efficient techniques used to classify, detect and track objects/drones. It uses efficient and state of the art machine learning techniques:

5.7 Robust Drone Classification Using Two-Stage Decision Trees and Results from SESAR SAFIR Trials [34]

The article proposes a two-stage decision-tree classifier to discriminate drones from other confuser targets. An Aveillant Gamekeeper 16U staring radar system produces raw data from which a set of features is extracted. The raw data consists of 4-D matrices in range, azimuth, elevation and Doppler for each frame. It goes through a step of CFAR (Constant False Alarm Rates) thresholding to produce positions of detections that are then assigned to different Kalman Filter generated track IDs. The features are extracted for each track at each frame and are related to kinematics or to the Doppler spectrum. The first stage of the tree is "hand-crafted" based on simple rules that use kinematics data derived from the tracker output of the staring radar, and the second stage is learnt with supervised learning. The structure of this tree aims at filtering some targets that are clearly not drones. The height feature can then be excluded from the second stage in order to prevent over-fitting because of its under-representation. Each track generated by the radar has a unique track ID. The track ID associated with the GPS truth of the drone is labelled as drone and all other tracks are assigned the label non-drones. The amount of non drone data is reduced to maintain a ratio of 3 to 1 with the drone data. For testing,the results were compared with those of a single stage decision tree classifier: the one-stage decision tree produces a True Positive (TP) ratio of 81.83% with a corresponding False Positive (FP) ratio of 0.87% for the two-class case. With the two-stage decision tree using the same training and test data set the results are 85.02% and 0.37% for TP and FP respectively.

5.8 YOLOv1 to YOLOv4: Optimal Speed and Accuracy of Object Detection [60, 62, 64, 5]

One of the current state of the art for real-time object detection that is recently commonly applied to drones detection and classification. Similarly to the previous versions, the system predicts bounding boxes using a potential set of bounding boxes that have been obtained through clustering. The network predicts 4 coordinates for each bounding box corresponding to offsets between the box and the final prediction. The network outputs an "objectness" score for each box, representing the probability that this box contains an object more precisely than any other. The class prediction is done in parallel for all the bounding boxes, using a feature extractor called a backbone network that can be easily switched. This backbone network comes from a classification network trained on the ImageNet data set. The one used for YOLOv4 is CSPResNext50 whose structure comes from DenseNet [30]. Tracking is implemented by Kalman Filters and comes after the drone detection score is high enough (a threshold is preset).

Open source implementation: https://github.com/AlexeyAB/darknet

5.9 EfficientDet: Scalable and Efficient Object Detection [84]

It is a Neural Network architecture that is more focused on reducing the number of floating point operations than YOLO4. It uses EfficientNet [83] as backbone network, whose structure is determined by neural architecture search. More specifically, a grid search is performed on a reduced convolutional neural network, its main building block being mobile inverted bottleneck layers [71][82], used in MobileNetV2. The resulting architecture is then scaled in a principled way, to produce a full scale architecture, able to get state of results on the ImageNet data set [70].

Open source implementation: https://github.com/xuannianz/EfficientDet

5.10 A Comparison of Convolutional Object Detectors for Real-time Drone Tracking Using a PTZ Camera [55]

The article compares the following six detectors whose tasks are followed by Kalman Filters for the purpose of tracking: YOLOv2[61], SSD [48] with MobileNet[29], SSD with Inception V2[31], R-FCN [12] with Resnet 101 [24], Faster R-CNN [66] with Resnet 101, and Faster R-CNN with Inception Resnet [80]. The data set has been produced by filming 11 different drones of varying sizes and models in different speeds, different distances and different background conditions. Images were then extracted from the videos and labeled manually. The tracking system used simply readjusts the rotation and the zoom of the camera in order to center the drone with a certain size. Accuracy and speed results show that Yolov2 is the best trade-off. However the article was published in 2017 so these detectors are quite old and it would be interesting to repeat the same procedure with current state of the art detectors, Yolov4 for example.

5.11 The Unscented Kalman Filter for Nonlinear Estimation [90]

An Extended Kalman filter that propagates estimates using a first-order linearisation of the transition and/or sensor models is presented. Clearly there are limits to such approximation, and in situations where models deviate significantly from linearity, performance can suffer. The unscented transform (UT) provides an alternative approximation. It characterises a Gaussian distribution using a series of weighted samples, sigma points, and propagate these through the non-linear function. A transformed Gaussian is then reconstructed from the new sigma points. This forms the basis for the unscented Kalman filter (UKF).

5.12 Lévy State-Space Models for Tracking and Intent Prediction of Highly Maneuverable Objects [15]

The article presents a Bayesian framework for maneuvering object tracking and intent prediction. Drone tracking is a potential application. The dynamical state of the tracked object is supposed to vary as a stable Lévy model:

$$dx(t) = Ax(t) + Bdt + HdW(t)$$

where H is a diagonal matrix and $W(t)$ is the isotropic multivariate α-stable Lévy process. The authors manage to approximate $x(t + \Delta t)$ as a truncated Poisson series summed with a Gaussian approximation of the residual. They also show how to use this model for inference using an efficient Rao-Blackwellized particle filter [14]. In contrast to conventional Gaussian formulations, this model is able to capture sharp changes in the state which may be induced by sudden maneuvers, making it probably appropriate for drone tracking. Results are presented on intent reference for human-computer interaction and for the tracking of a maneuvering vessel. The comparison with a standard Gaussian model shows that indeed this model is better at capturing the abrupt changes of behaviour.

5.13 A Neural Network Adapted Kalman Filter (NNAKF) for Target Tracking [37]

This articles proposes a new way to augment Kalman filters with a neural network in order to track maneuvering targets in 3 dimensions. A common practice is to assume a constant velocity for the target and to adapt the process noise covariance matrix through time. If its coefficients are set too low, the filter trusts the model and thus obtains good performance when the actual velocity is close to being constant but degrades when maneuvers are performed, whereas if they are set too high, the filter trusts more the radar measurements, which is relevant during maneuvers but leads to degraded performances in straight lines (Fig. 1). Adapting this matrix through time had first been suggested by Castella [9] with a simple procedure. This procedure inspired the structure of the new filter represented in Fig. 2, where, in general, the X_i represents the state variables values, P_i is the covariance of the state variables, Q_i is the process noise,I_i is the innovation (which is the difference between measured values and updated measured values), Y_i are the measurements, and R_i is measurements noise. Please refer to[37] for more details. The covariance matrix results from a linear combination of matrices $\sum_{i=0}^{N} \sigma_i Q_i$ where N is a hyper-parameter, these matrices consist of trainable parameters and the coefficients σ_i are the outputs of a (Long Term Short Term Memory) [26] neural network, allowing it to remember past information. The inputs of this network are the normalized square innovations for each axis. To evaluate the filter, artificial trajectories are simulated from random Gaussian flight commands and smoothed by a low-pass convolution filter. The performance is compared with other filters such as:

- Regular Kalman filter using a constant speed model
- Castella: this filter adds a simple adaptation to the kalman filter with constant velocity model using the normalized square innovations to add noise on the corresponding speed component. Only a few more parameters are added to the kalman filter to tune this adaptation.
- Vaidehi [87]: this variation of the kalman filter proposed by Vaidehi uses a neural network that takes the innovation, gain and correction to produce an additional cor-

rection to the estimation provided by a kalman filter with constant velocity model. The original filter was made to track four targets at once, but can easily be adapted to a mono-target problem.
- LSTM: this filter replaces the evolution process by a neural network.

NNAKF is shown to significantly outperform these four other filters on the generated trajectories.

5.14 Invariant Extended Kalman Filter Applied to Tracking for Air Traffic Control [51]

This article applies an Invariant Extended Kalman Filter (IEKF) to track planes in the context of air traffic control. The target model is expressed using the Frenet-Serret frame, in which it is possible to represent any 3D trajectory. The frame is entirely determined by the three following vectors: the tangent vector T, the normal vector N and the bi-normal vector B, all represented in Fig. 3. The evolution of these three vectors is expressed with u, the norm of the velocity of the center of the frame, κ, the curvature of the trajectory and $\tilde{\tau}$, the torsion of the trajectory.

$$\frac{d}{dt}T = u\kappa N, \frac{d}{dt}N = u(-\kappa T + \tilde{\tau}B), \frac{d}{dt}B = -u\kappa N$$

Two hypothesis are made in order to derive a model on which to apply the invariant Extended Kalman Filter:

- The target does not slide, meaning that the velocity vector of the target and the tangent vector T of the Frenet-Serret frame are always co-linear,
- The curvature, the torsion and the norm of the velocity u of the target are nearly constant.

The performance of the new algorithm is compared with the one of an Extended Kalman Filter (EKF) applied to a constant velocity target model. The experiments are performed in a realistic simulation environment, modeling the entire tracking chain algorithms. Several trajectories are simulated, for different types of scenarios: fully radial (with constant velocity), fully tangential (also with constant velocity), or performing a u-turn. The IEKF outperforms the EKF in position and velocity precision (measured with a root mean squared error), for all types of trajectories.

5.15 Maneuver Detector for Active Tracking Update Rate Adaptation [57]

This work proposes a new method to detect maneuvers from a target, in order to order urgent pointing with a wider beam width, when necessary, to avoid target loss. As for [51], a Frenet-Serret Model is considered and the tracking is done with an Invariant Extended Kalman Filter. However this time, it is based on the assumption of piece-wise constant velocity's norm, curvature, and torsion, with jumps of those parameters at unknown random times. To detect the maneuvers, a variable rate particle filter is used. This filter is extensively described in [19]. The (Variable Rate Particle Filter) VRPF predicts these jumping times, with the hypothesis that the jumps can occur at any moment, unlike the jump Markov linear Gaussian model of [15], which consider a discrete-time model where jumps and output measurements are synchronized, which leads to algorithms such as the Rao-Blackwellized particle filter [14].

This particle filter is used in coordination with the IEKF. More particularly, the IEKF estimates are used to initialize a sliding window VRPF that lives on for a few steps and that is responsible to estimate the probability of a jump occurring.

The detector is shown to be working on an example trajectory with two sudden changes of parameters. Its computation cost is small enough and it can easily be parallelized.

5.16 An Algorithm for Large-Scale Multi-target Tracking and Parameter Estimation [8]

In contrast to the previous articles, this one focuses on tracking a high number of targets at the same time. Rather than modeling the dynamics of one object, point processes are used to describe a stochastic population of targets. Multi Target Tracking (MTT) techniques are presented here with the application of Kalman Filters as state estimators.

5.17 Deep Kalman Filters [42] (2015)

This paper introduces a method to perform state estimation by using generative neural networks for the transition and the observation function, in association with a recognition network. These networks are trained with stochastic back-propagation, by maximizing a lower bound on the likelihood of the model given the data set, a technique that has been developed for variational inference. Different architectures for the recognition network are compared, some using a recurrent neural network or a bi-directional recurrent neural network. The model is evaluated on the "Healing MNIST" data set, which contains perturbed, noisy and rotated MNIST digits. They also use it for counterfactual estimation using electronic health records from 8000 diabetic and pre-diabetic patients. In both cases these applications are quite far from drone tracking because the number of dimensions is higher.

5.18 Online Tracking and Re-acquisition Using Co-trained Generative and Discriminative Trackers [91]

This article proposes two co-trained models for tracking purposes, one generative and one discriminative. The model was based on classical machine learning, SVM in particular.

5.19 Backprop KF: Learning Discriminative Deterministic State Estimators [22] (2017)

In contrast to the previous article that trains a generative model, this one focuses directly on training a discriminative model as a unique deterministic computation graph that can be trained with back-propagation. This corresponds to a type of recurrent neural network model, where the architecture of the network is informed by the structure of the probabilistic state estimator, in that case the one of a kalman filter, as shown on Fig. 4. The graph is composed of a feed-forward part, which processes the raw observations o_t and outputs intermediate observations z_t and a matrix L^t that is used to form a positive definite observation covariance matrix R_t, and a recurrent part that integrates z_t through time to produce filtered state estimates.

One advantage of this approach is the ability to incorporate arbitrary nonlinear components into the observation and transition functions. The method is evaluated on a data set of images with the task of tracking a disk on a sequence of coherent images. It shows better results than naive neural network based estimators.

5.20 Box Particle Filtering for Nonlinear State Estimation Using Interval Analysis [1]

This article uses interval analysis to replace the points of a classical particle filter, with box particles. In contrast to kalman filters that make Gaussian assumptions, interval analysis suppose that uncertainties are bounded and updates the bounds of the intervals with each new information. The model is experimented for the localization of a vehicle using GPS, a gyro and an odometer.

5.21 H_∞ Adaptive Filtering [23]

This classical filter is the first of an entire class of filters oriented towards robustness regarding noise, disturbances or non cooperative control inputs. It applies to linear models but as with extended kalman filters, local linearizations can sometimes be used. The performance of the filter is compared to the one of a H_2 filter that uses a quadratic criterion instead of a worst case criterion (Table 1).

A comprehensive review of methods for state estimation, from classical model-driven methods to data driven methods using machine learning has been presented so far. Table 2, summarizes the different tracking filters that have been developed until recently. Some of them are used in the work presented above.

Table 1 Speed Vs. Accuracy comparison on VOC2007 (State of the art in 2015)

Method	Speed	Accuracy (mAP)
Faster R-CNN	7 FPS	73.2 %
YOLO	45 FPS	63.4%
SSD	59 FPS	74.3%

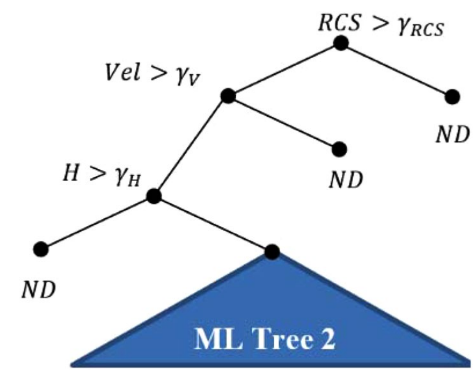


Fig. 1 Two stage decision tree. RCS, Vel and H are features for target radar cross-section velocity and height, respectively. YRCS, YV, YH are the feature threshold values that are set a-priori in stage one tree

6 An Analysis Case: Regression Using Nonlinear Polynomial Method (NPR) Versus Linear Kalman Filter (LKF) Tracking

The tracking data for a moving object in two dimensions is investigated in this section. Tracking is implemented based on Table 3 below that provides a possible ground truth for a track. The first column represents the location and the second column represents the velocity in two dimensions. First, the trajectory of the track is estimated based on nonlinear regression techniques [67] using Least Square Methods (LSM) for residuals minimization.

For simplicity, tracking is implemented for one location variable which is x versus time. The trigonometric regression was applied. Figure 5 shows the tracking trajectory based on the values given of Table 3. The trigonometric regression applied follows the following equation:

$$y = a\sin(x - b) + cx^2 + d \tag{1}$$

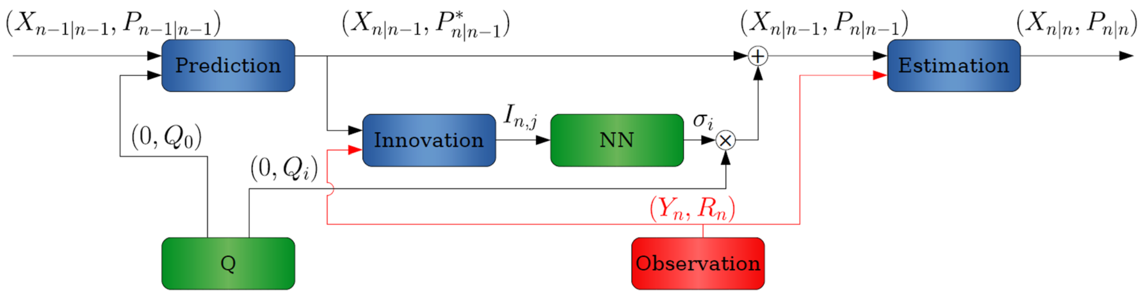


Fig. 2 NNAKF

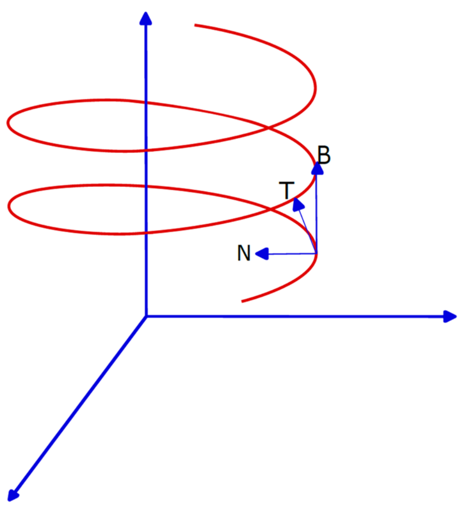


Fig. 3 Frenet-Serret frame

By inspection, it is clear that trigonometric regression is not fairly close to the ground truth points as it has a cyclic nature and does not exhibit flat and then sharp turns that a drone my take (Figs. 6, 7, 8, 9, 10, 11, 12). Therefore, more complex polynomial non-liner regression (PNR) was also applied. Below is a PNR form with degree = 7:

$$y = ax^7 + bx^6 + cx^5 + dx^4 + ex^3 + fx^2 + gx + h \tag{2}$$

PNR was applied for tracking the given values in Table 3. Plots for regression of polynomial degrees from 7 until degree 13 were generated. It has been noticed that the summation of the absolute value for the residuals were decreasing as the polynomial regression degree was increased from 7 until 13, where the sum of absolute residuals becomes close to zero. After that, it is easy to notice the over-fitting, where the curve can take broken shapes. The broken shape can physically represent light and high maneuvering drones. However, the ground truth points in our example resemble an extreme case that is used to test how good the PNR can be.

Table 4 and Fig. 13 show the sum of absolute residuals with respect to the degrees of the polynomials used in regression. Apparently those quantities reflect the closeness of the curves generated with the respect to the ground truth points. The quality of the generated curve can be verified visually and quantitatively based on the sum of the residuals.

On the other hand, Linear Kalman Filter (LKF)is used to estimate the state of dynamic systems. The LKF has two assumptions:

1. The world is Gaussian.
2. All models are linear.

Stone Soup https://stonesoup.readthedocs.io/en/v0.1b5/ was used in simulating LKF. For more details on LKF, Extended Kalman Filter (EKF), and Unscented Kalman Filter (UKF) please see [53]. The analyses start by testing the effect of changing the measurements noise co-variance on the tracking results. The goal was to set the first co-variance value to 0 and then increase it step by step 0.2 until the value 10

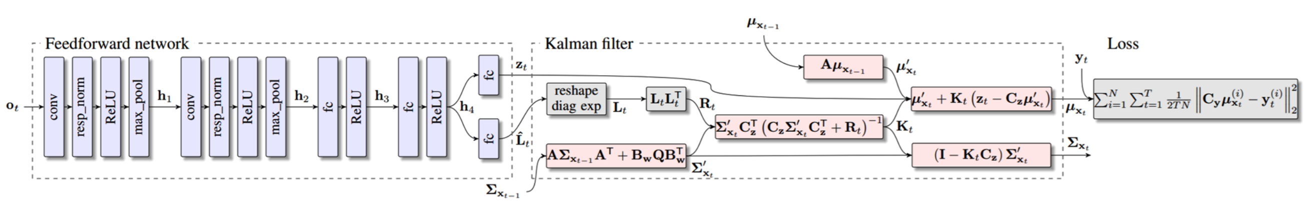


Fig. 4 Computation graph of Backprop KF

Table 2 Comparisons between different filters including KF types

Filter	Requirements for the System	Accuracy for a Practical System	Calculation Cost	Description
Linear Kalman filter	Linear, with Gaussian white noise	Low	Low	The requirements for the system are very high, so it is difficult to achieve high accuracy in the actual application system
EKF(Extended KF)	Nonlinear, with Gaussian noise	Medium	Low	*covered above*
UKF(Unscented KF)	Nonlinear, with Gaussian noise	Medium	Medium	*covered above*
CKF(Cubature KF [72])	Nonlinear, with Gaussian noise	Medium	Medium	The performance of UKF and CKF is better than that of EKF, but their calculation amount is slightly larger than that of EKF
Gaussian mixture filters [89]	Nonlinear, with non-Gaussian noise	Medium	Medium	A recursive algorithm proposed for jointly estimating the time-varying number of targets and their states
Particle filters [21]	Nonlinear, with non-Gaussian noise	High	High	These filters have low requirements for the system. However, the amount of calculation is large

Table 3 Tracking data

Time	Location(x,y)	Velocity(Vx,Vy)
1	(1,1)	(4,5)
2	(1,2)	(5,6)
3	(1,3)	(5,5)
4	(1,4)	(5,5)
5	(2,5)	(7,6)
6	(2,6)	(4,6)
7	(3,5)	(5,7)
8	(3,6)	(6,8)
9	(4,6)	(7,7)
10	(4,7)	(8,7)
11	(4,8)	(9,8)
12	(4,9)	(7,9)
13	(4,10)	(9,9)
14	(4,11)	(6,7)

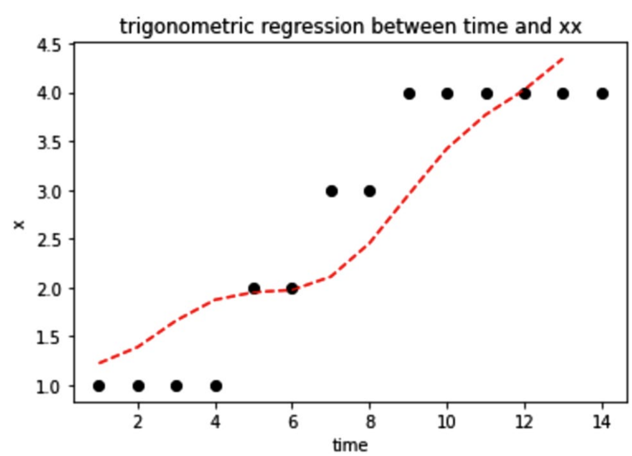


Fig. 5 Trigonometric regression for the relation between time and x

is reached. It was observed that the increase in uncertainty (circles in orange of Figs. 14 and 15) as the measurements noise co-variance grows. For better illustrations Figs. 14 and 15 show some tracks and measurements values for two different ground truths.

The second step was to find the summation of Euclidean Distances between a ground truth path and the tracking curve. As observed, when the noise co-variance increases the distance values also grow. The values are shown in Table 5:

Below are some Figs. 16, 17, and 18 that show the tracks generated by LKF compared to the tracks generated

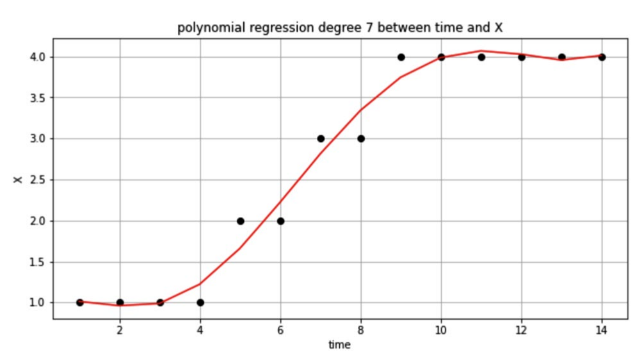


Fig. 6 Regression with degree 7 for the tracking data

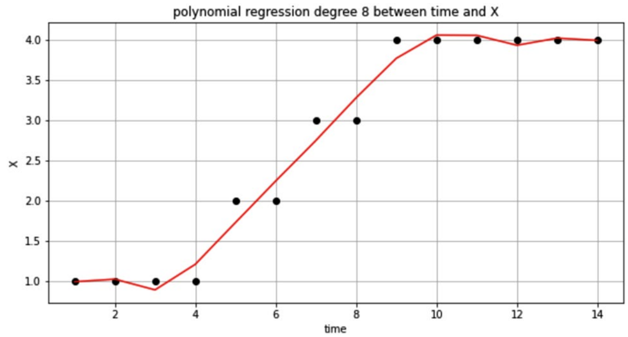


Fig. 7 Regression with degree 8 for the tracking data

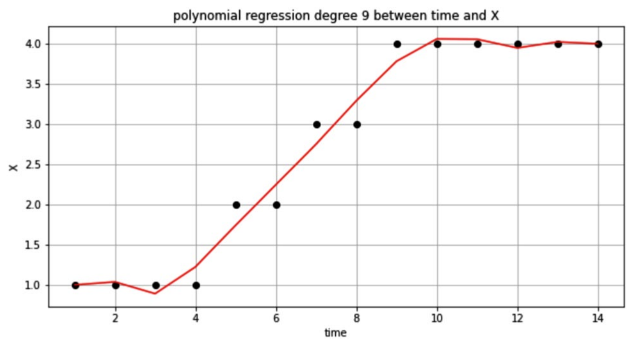


Fig. 8 Regression with degree 9 for the tracking data

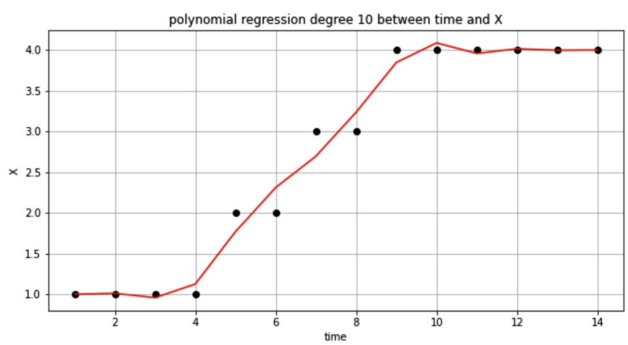


Fig. 9 Regression with degree 10 for the tracking data

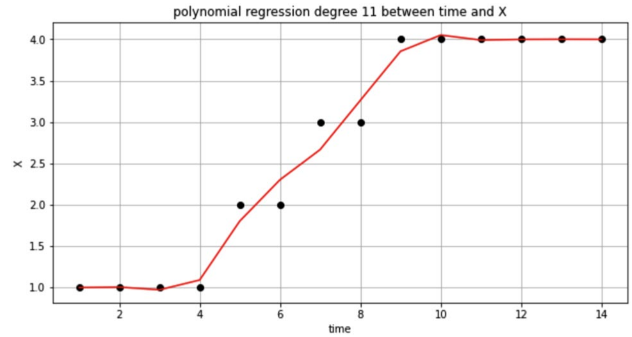


Fig. 10 Regression with degree 11 for the tracking data

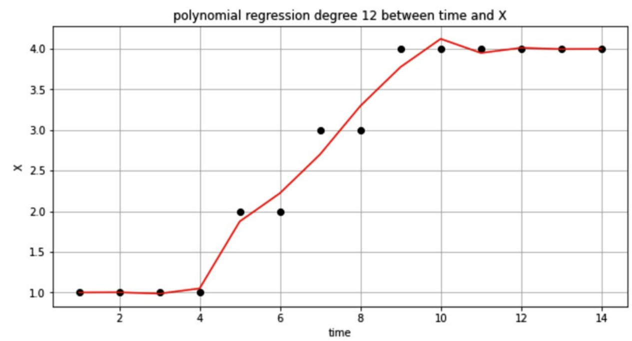


Fig. 11 Regression with degree 12 for the tracking data

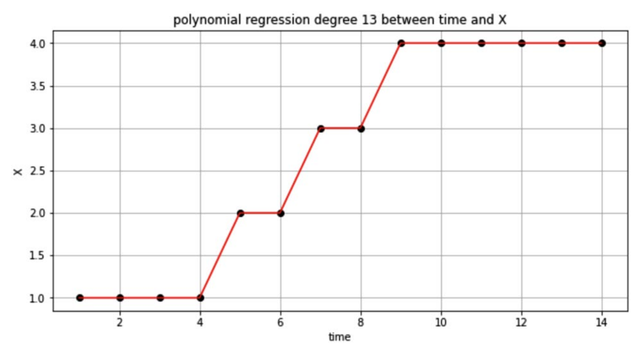


Fig. 12 Regression with degree 13 for the tracking data

by nonlinear polynomial regression (NPR) with different ground truth and different noisy measurements. The NPR here is applying regression on the measurements values simulated by Stone Soup.

This time, the sum of the Euclidean Distances between the ground truth values and the generated tracks values were repeated with one identical ground truth in all simulations. Different covariance values for the noisy measurements were used while different degrees for the NPR are also used for regression. Regression was applied on measurement values. The results are shown in Table 6:

For more comprehensive analysis, a total of 10 different randomly generated ground truths were used. Comparisons between the LKF and the NPR were done based on averages of the summation of Euclidean Distance values for the 10 ground truths. The LKF tracking was applied with different co-variance values from 5 to 10. Different degrees of nonlinear polynomial regression (from n=1 to 5) were also tried. Regression was applied on measurement values. The results are shown in Table 7:

The average Euclidean Distances in Tables 6 and 7 decreased rapidly as the dimensions of the polynomial

Table 4 Degree Vs summation of the absolute values of residuals for the tracking data

Summation of the residuals	Degree
4.897	1
4.308	2
2.736	3
2.276	4
1.870	5
1.783	6
1.797	7
1.836	8
1.824	9
1.562	10
1.424	11
1.421	12
1.336×10^{-05}	13
4.296×10^{-06}	14
1.664×10^{-06}	15
4.509×10^{-07}	16
8.174×10^{-07}	17
2.138×10^{-07}	18
2.914×10^{-07}	19
6.518×10^{-08}	20

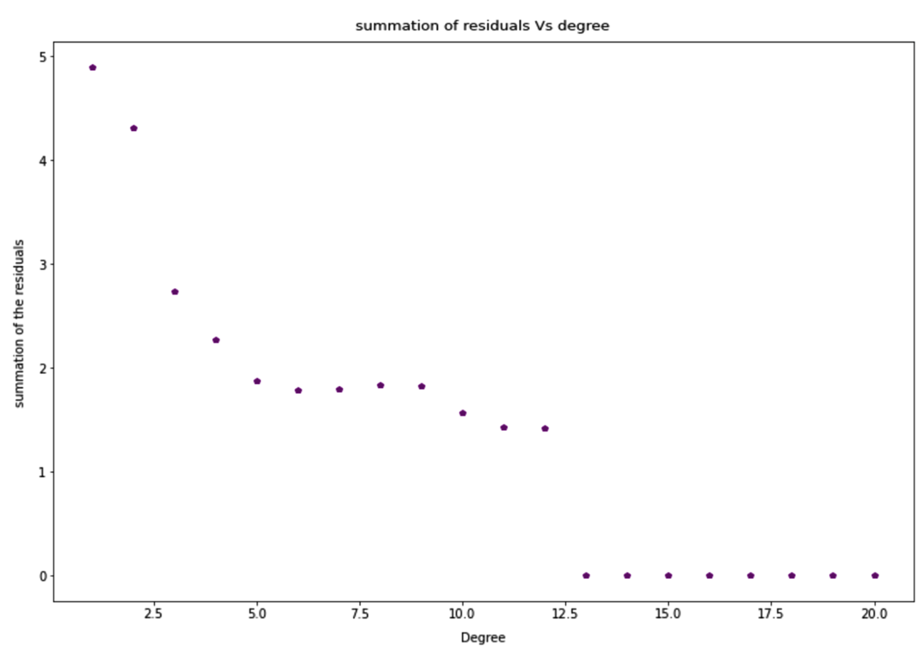
increased in the NPR case. In the LKF case, and from the tables we noticed that the Euclidean Distance error increased as the co-variance increases. However, the results of a single ground truth of Table 6 can not be relied on to reach a conclusion. For such type of simulations that have a stochastic nature it is recommended always to have a Monte Carlo type of simulations similar to the ones we had in Table 7. It is evident that Linear Kalman filter (LKF) outperformed the NPR method in this type of noisy environments. The NPR deals directly with the measurements that could be noisy and deceiving. The NPR estimation showed more resilience and less sensitivity for the covariance changes. However, based on the Euclidean Distance measures the LKF outperformed the NPR. Without using a motion model that can help in predicting the next state as LKF does, it is harder for the NPR to estimate well the tracks that that have high levels of noisy measurements.

7 Conclusions and Discussions

In this work, plenty of literature on tracking and detection was presented. The techniques are different in methodologies and heuristics. Most of them take uncertainty into consideration by using probabilistic and statistical based models to estimate the different variables; either of the physical object or the measurements. It has been found that Kalman Filter(KF) based techniques are still among the most used and many researchers still use them in tracking the objects. AI-based techniques can be blended or incorporated within the KF models to enhance the performance and add more adaptability to the tracking process. AI-based techniques are successful in detection, recognition, and classification processes.

Another insight from [81] is that the radar based models have on average better performance compared to visual (>90% against>80%). Indeed [2, 46, 58, 85] demonstrate how different features extracted from objects spectrograms can help efficiently classify them. It is possible to get speed,

Fig. 13 Sum of the absolute values of residuals Vs Polynomial degree for the tracking data



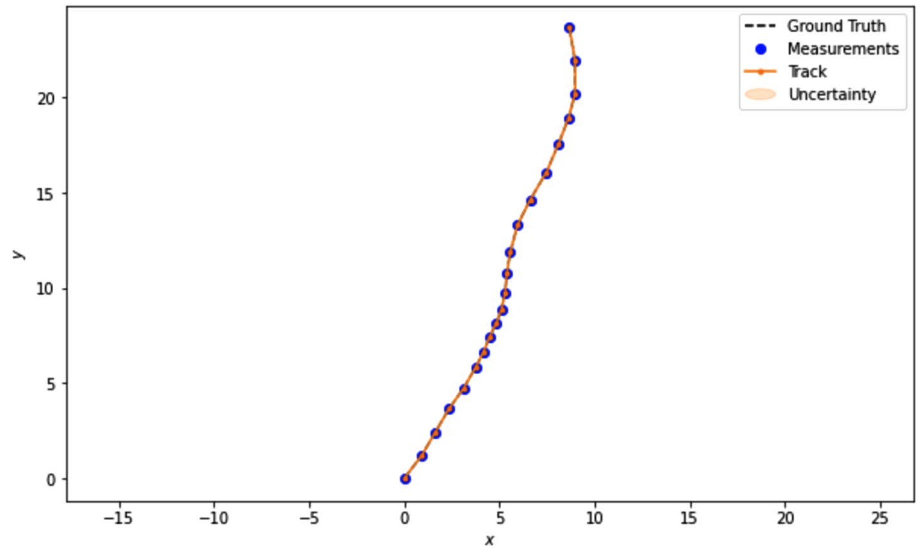


Fig. 14 The tracking line and the truth path with co-variance 0

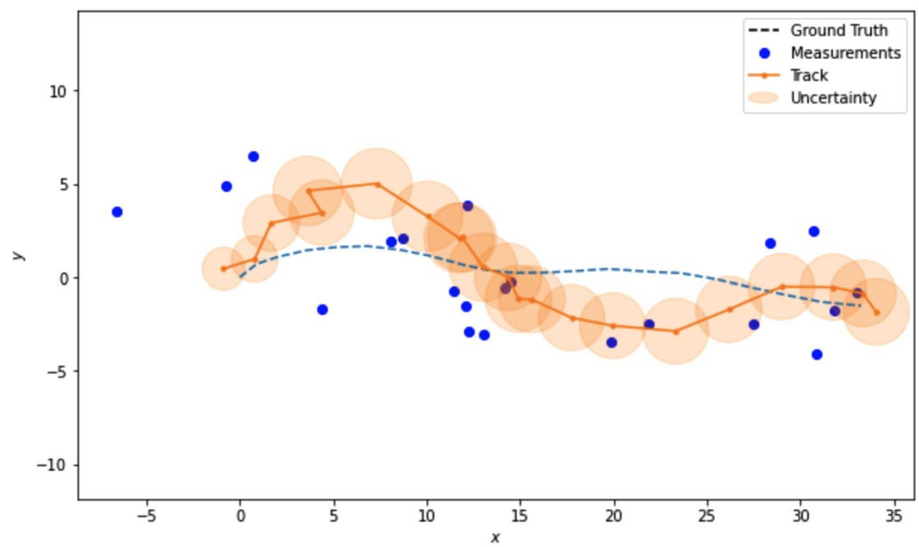


Fig. 15 The tracking line and the truth path with co-variance 10

Table 5 The Euclidean Distance between Tracks and Ground Truth with different noise measurement co-variances

Summation of the Euclidean Distance values	Co-variance
0.0	0
16.083	1
16.828	2
22.140	3
33.010	4
37.239	5
40.246	6
37.690	7
41.167	8
38.465	9
44.221	10

acceleration, heading, wavelengths and other features that can be turned into feature vector and be classified using SVM or other classical methods. All these papers are done using purely radar frequency data sets. Therefore it would be logical to perform detection using radar data. If the research work has to deal with videos, we need to find a way to transform it into radar spectrograms. There is only one work done in such area. [3] demonstrate how videos of human activities can be transformed into spectrograms. If the same technique could be applied to drone detection this would be a breakthrough. However, a lot of limitations will not allow this. First of all, there is no data set of both drones and their radar frequencies, which is essential for [3]model. Even if there was a chance to collect this data, the videos must be of single drones, using static camera and without any other

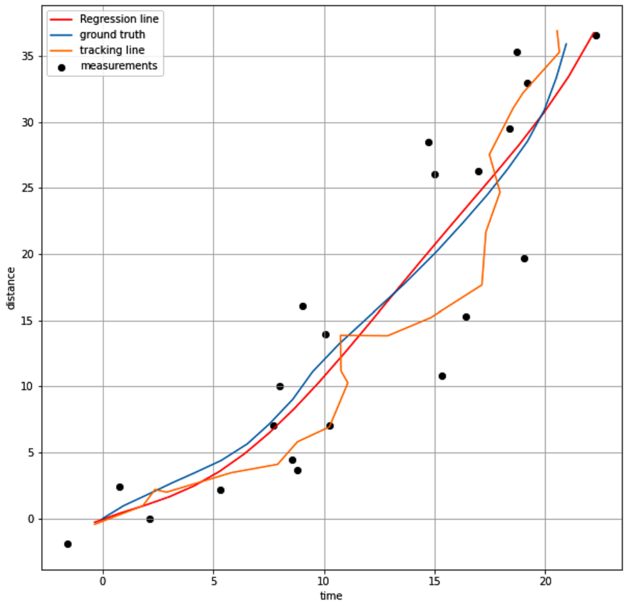


Fig. 16 The tracking with LFK and nonlinear regression- degree=3, cov=5

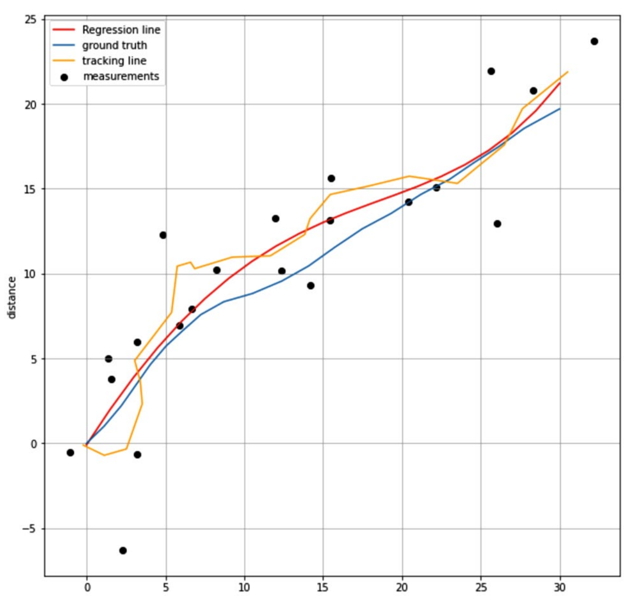


Fig. 17 Regression line of degree 4 with co-variance 6

moving objects. This method is very unlikely to perform well on complicated test data.

For visual techniques, in particular, Deep Learning (DL) methods, there is a possibility to extract histograms of gradients from the image [50, 73]and use it in K-Nearest Neighbour (KNN), Random Forest (RF) or SVM directly to classify. However, [50, 68]show that classical methods lack capturing high-level details and fail to identify small

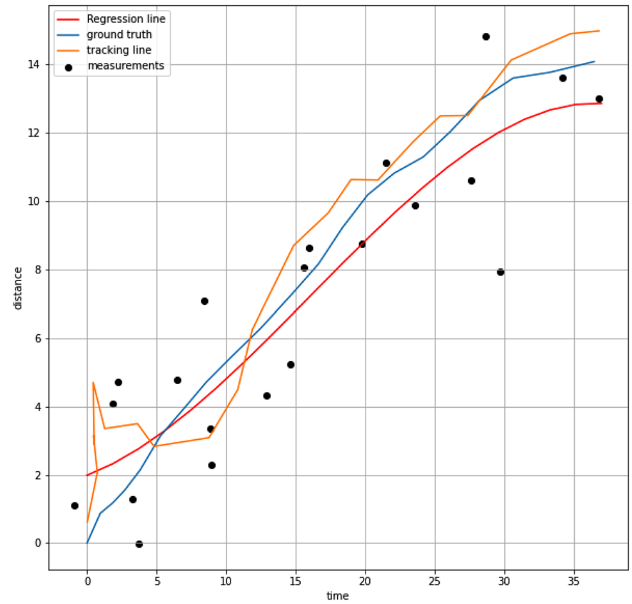


Fig. 18 The tracking with LFK and nonlinear regression- degree=5, cov=10

targets. This leads to usage of deep learning models in drone detection. There are two approaches in this field: single-stage and two-stage models. In single-stage, the models are able to detect and classify the object using one network, in two-stage it is mostly using some DL model for detection and then classification with simple models. It is unclear which approach performs better. [32, 93]show that two-stage methods perform better (especially Faster-RCNN), but [11] demonstrates one-stage models success compared to other. However, thorough research shows that performance really depends on which data and what kind of modifications are applied to the networks. In order to improve the results mentioned before, it is a must to know the pain-points of this area (i.e. lack of data area). Not only there is a lack of drone data sets, but there is also certain requirements such as having moving camera, multiple objects, diverse backgrounds, and etc. Additionally detecting drones in the nighttime requires infrared images, which probably should be addressed in all related research work. It is also agreed that synthesizing data sets is an inevitable part for drones detection and tracking. There are several ways of doing so. Both visual and radar tools for data collection become limited after 500 m distance as shown in [49, 52]. Moreover, Faster-RCNN (Region Based Convolutional Neural Network)and couple other models reduce image size in their networks which affects small object detection (e.g. objects of size 16x16 becomes 1px in Faster-RCNN [7]). Evidently, researchers are in need to apply super-resolution models as [10, 49] did. There are many other models that are based on transformers, GANs (Generative Adversial Models) and

Table 6 The summation of Euclidean Distances using NPR and Kalman Filter with different covariances and different NPR degrees (shown for single cases and average). One ground truth only is used

Co-variance						
Degrees	5	6	7	8	9	10
1	23.41	23.41	23.41	23.41	23.41	23.41
2	21.54	21.54	21.54	21.54	21.54	21.54
3	21.34	21.34	21.34	21.34	21.34	21.34
4	21.94	21.94	21.94	21.94	21.94	21.94
5	24.25	24.25	24.25	24.25	24.25	24.25
Average(Regression)	22.49	22.49	22.49	22.49	22.49	22.49
Linear Kalman Filter	20.88	23.90	18.85	38.28	28.60	25.24

Table 7 The Average of the summation of Euclidean Distances using NPR with different degrees and covariances /and for Kalman Filter with different covariances (Averages are taken over 10 different ground truths)

Co-variance						
Degrees	5	6	7	8	9	10
1	84.49	36.70	35.93	40.84	43.57	40.43
2	20.83	45.89	23.78	15.15	19.51	27.73
3	15.90	42.90	43.85	18.25	12.51	27.63
4	15.87	42.12	47.29	28.03	16.09	31.57
5	13.81	42.73	46.64	27.23	16.14	29.43
Average(Regression)	30	41.6	39.4	25.88	21.5	31.35
Linear Kalman Filter	19.37	22.96	27.76	25.00	26.05	21.59

etc. Their performance varies slightly and the most common metric has been criticized a lot. If data is synthesized as additional training/testing data, it must be much less in size than the real data. The same techniques can be used to generate infrared images as [4]did. Another extension of nighttime data used for drones detection and tracking can be done by converting RGB images to gray-scale and tweaking it to look like infrared images.

On the other hand, the analysis implemented in this paper covered a deterministic approach which is the nonlinear polynomial regression (NPR) approach that assumes measurements have no covariance. We made simulations for tracking using the NPR approach. The Least Mean Square (LSM) algorithm was used in minimizing the residuals [67]. The interesting finding was that at certain dimensionality of the polynomial, the fitting curve (track) tends to become over fitted and the track would appear unrealistic. It is healthier to choose tracks with some minimal value of total residual error otherwise they will appear like zigzag and unrealistic. The simulations with Linear Kalman filter (LKF) were implemented using stone soup platform. The LKF, as a matter of fact, uses three covariance matrices; the model, the process, and the measurements. On the other hand the NPR was completely deterministic. However, both were using the same measurements. The NPR did not use any model for the object, while the LKF had to use a simplified motion model with constant speed. One variable, the horizontal location, was traced for simplicity and for graphical feasibility. Figures 16, 17, and 18 show samples of the tracking results under different co-variances and ground truths. Apparently the NPR tracks show more conservative behavior than the LKF tracks. The LKF uses all the information embedded in the physical model, the covariance matrices, and the measurement values to estimate the current state of the object. This is unlike the NPR that only uses the measurement values. Consequently, and as shown in the mentioned figures, the trajectory generated by LKF is more sensitive to the variations of the sparsely collected measurements and shows more interaction with their random variations along the ground truth. Table 7 shows comprehensive and direct comparisons between the Euclidean distances (average of sum of Euclidean distance for every single time step of 5 s) for the NPR and the ground truth from one side and between the LKF tracks and the ground truth from another side. Apparently, higher degrees of NPR result in less Euclidean Distances (ED) values regardless of the co-variance level. Of course, higher covariance values result in higher ED values for all degrees of NPR. However the NPR showed less sensitivity to those changes. The most important finding is that the values of ED were, in general, less when using LKF for tracking. Looking at the values in the last two rows of Table 7 supports this fact except in one case when measurements covariance value was 9. This could be due to the stochastic behavior of the measurements sampling at that very case. The simulations were done using different and random ground truth at every case of study. As mentioned earlier we had 10 different cases. This is done to eliminate the effect of “ground truthing” when comparing between the two tracking mechanisms. Table 6 shows the comparisons between the ED values for both the LKF and the NPR with the same and one ground truth. Apparently, fixing the ground truth has made the ED values of the NPR constant for the same degrees at different covariances. It shows that the covariances variations did not affect the regression outcome. On the other hand, there is an overall increase in the ED values at higher covariance values with the LFK method. Therefore, it was necessary to rely more on the outcomes of Table 7 in which 10 different and random simulation were implemented. Single case (one ground truth) simulations as in Table 6 can not be conclusive.

When applying tracking for a moving target, it is advisable to start with LKF. The LKF clearly outperformed the NPR in our simulations. However, the NPR has less requirements as it uses the measurements model only and does not require a motion model, while the LKF requires both a motion model and a measurement model. In some cases it is hard to build a an acceptable motion model, specially when the motion model is highly nonlinear. The LKF assumes linearity of the motion model and in that case it would be rational to rely on NPR to avoid errors that could be caused by using inappropriate linearized motion model. As the motion tends to become linear, the LKF can be invoked to handle tracking since it has better over all performance.

Finally, the paper provides a sufficient review of drones tracking and detection. It is accompanied by detailed analyses using two simplified, yet vastly used, techniques of tracking and regression. Several open sources were used and referred, in addition to a large number of references necessary to support any research in that field. Future work will include more detailed revisions and analysis for radar based techniques accompanied with machine learning methods used for drones/birds classification.

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